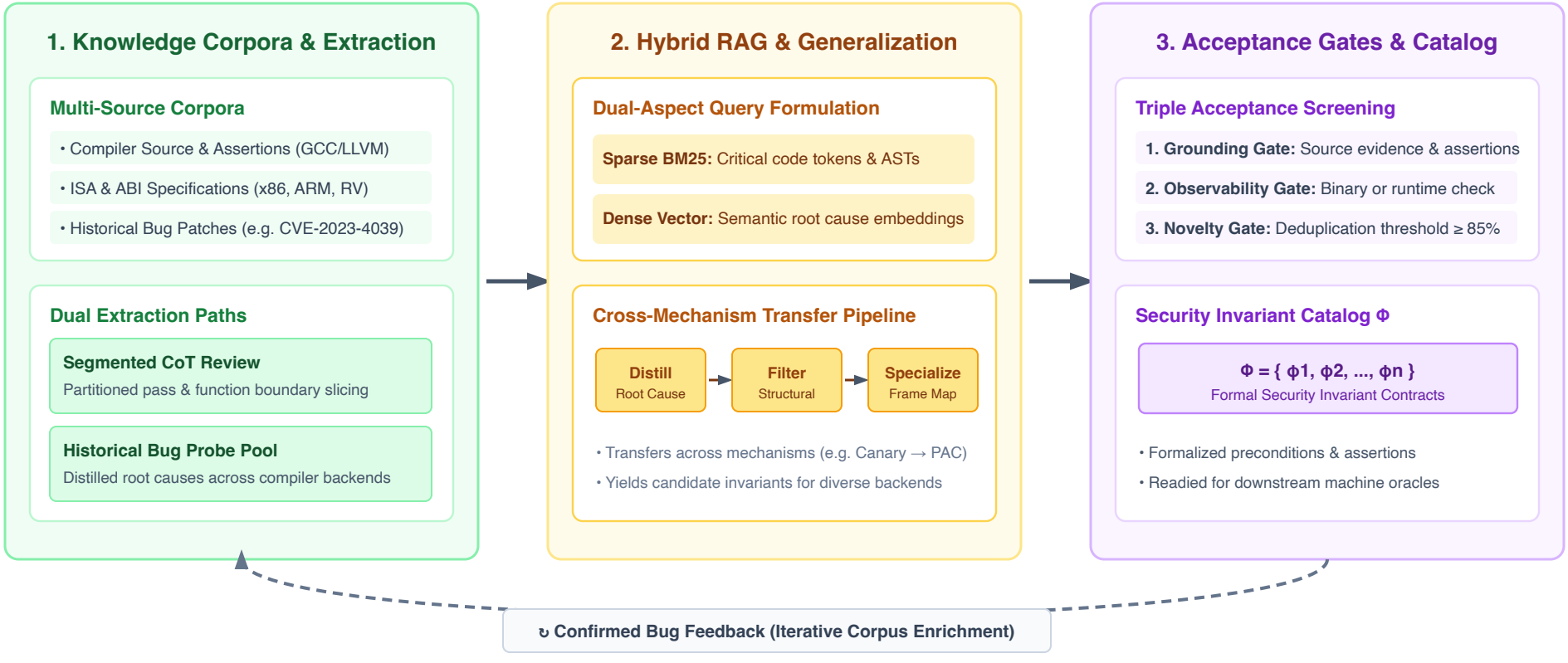

1. Knowledge Corpora & Extraction
Multi-Source Corpora
• Compiler Source & Assertions (GCC/LLVM)
• ISA & ABI Specifications (x86, ARM, RV)
• Historical Bug Patches (e.g. CVE-2023-4039)
Dual Extraction Paths
Segmented CoT Review
Partitioned pass & function boundary slicing
Historical Bug Probe Pool
Distilled root causes across compiler backends
2. Hybrid RAG & Generalization
Dual-Aspect Query Formulation
Sparse BM25: Critical code tokens & ASTs
Dense Vector: Semantic root cause embeddings
Cross-Mechanism Transfer Pipeline
Distill
Root Cause
Filter
Structural
Specialize
Frame Map
• Transfers across mechanisms (e.g. Canary → PAC)
• Yields candidate invariants for diverse backends
3. Acceptance Gates & Catalog
Triple Acceptance Screening
1. Grounding Gate: Source evidence & assertions
2. Observability Gate: Binary or runtime check
3. Novelty Gate: Deduplication threshold ≥ 85%
Security Invariant Catalog Φ
Φ = { ϕ1, ϕ2, ..., ϕn }
Formal Security Invariant Contracts
• Formalized preconditions & assertions
• Readied for downstream machine oracles
↻ Confirmed Bug Feedback (Iterative Corpus Enrichment)